Annex B

Cover Page: Millennia Institute H2 Mathematics 9740

Qn/No	**Topic Set**	**Paper Number**	**Answers**	**Marks**
1	Inequalities	9740/01	$x < -2$, $-1.56 \le x \le -1$, $2 < x \le 2.56$	4
2	Method of Difference	9740/01	(i) $-\frac{2}{n}+\frac{2}{n-1}$ (ii) $1 - \frac{1}{N}$, 1	5
3	Equations	9740/01	(100 450 250), factory should stop producing White chocolate bars as it has the least profit margin.	5
4	AP & GP	9740/01	(ii) 0.5 (iii) $2a(1 - 0.5^n)$	8
5	Graphing Techniques	9740/01	(a)(i) $x = 4$, $y = 3$ (iii) $k < -1.66$ or $k > -0.339$	15
6	Differentiation	9740/01	(a) $2x - \frac{5x^3}{6} + \ldots$ (b) $-\frac{4}{25}\pi\ cms^{-1}$	9
7	Vectors	9740/01	(i) 10.1^o (ii) $\begin{pmatrix} 5/2 \\ 0 \\ 7/2 \end{pmatrix}$ (iii) $\mathbf{r} = \begin{pmatrix} 15/7 \\ -8/7 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 5/7 \\ 2/7 \\ -1 \end{pmatrix}$	12
8	Maclaurin's Series Differentiation	9740/01	(iii) $y = \frac{8}{81}x + \ln\left(\frac{3}{2}\right) - \frac{1}{3}$, 0.0263	9
9	Summation of Series	9740/01	(i) $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$ (ii) $\frac{1}{n}$	9
10	Complex Numbers	9740/01	(a)(i) $z = 2e^{i\left(\frac{-\frac{\pi}{2}+2k\pi}{4}\right)}, k = -1, 0, 1, 2$ (b) $a = -1.6$, $b = 4.2$	11
11	Integration & Differential Equations	9740/01	(a)(i) $\frac{\pi}{4}$ (ii) $\int e^x \sin x\, dx = \frac{1}{2}e^x(\sin x - \cos x) + c$ (b)(i) $y = \ln(x^2+1) + c$ (ii) $y = \ln(x^2+1)$ (iv) gradient tends towards zero	13
Qn/No	**Pure Maths**	**Paper Number**	**Answers**	**Marks**
1	Definite Integrals	9740/02	$\frac{5}{3}\pi$	5
2	Maclaurin's Series	9740/02	$y = -x + \frac{1}{2}x^2 - \frac{1}{12}x^4 + \ldots$	7
3	Definite Integrals	9740/02	---	7
4	Complex Numbers	9740/02	(i) 1, 3 (ii) $2 - \sqrt{3}\,i$	7
5	Functions	9740/02	(i) $f^{-1}: x \to -\sqrt{3-x}$, $x \le 3$ (iii) -2.30 (v) $(-\sqrt{3}, \sqrt{3})$ $\ln(3 - x^2)$, $x \in (-\sqrt{3}, \sqrt{3})$ $y \le \ln 3$	14
Qn/No	**Statistics**	**Paper Number**	**Answers**	**Marks**

6	Sampling	9740/02	too costly or time consuming Example should illustrate • correct strata size; • random selection from each stratum a good representative sample of the population (or equivalent)	4
7	Permutation & Combination	9740/02	(a) 5148 (b) 31 (c) 420	6
8	Hypothesis Testing	9740/02	(i) do not reject H_0, insufficient evidence… (ii) assume population is normally distributed	7
9	Binomial Distribution	9740/02	(a)(i) 0.904 (ii) 0.0120 (b) 0.210	8
10	Correlation & Regression	9740/02	(i)0.972, strong positive linear correlation (iii) y = 10.1x + 2.58 Slope represents expected increase of 10.1 dead cockcroaches for every 1% increase in the concentration of the insecticide. (iv) Not reliable as larger values of x are out of range so data may not be linearly correlated. (v)2.58	9
11	Probability	9740/02	(a)(i) P(A) = ½ , P(B) = 11/36, P(C) = 1/12 (ii)not independent (iii)5/6 (b)(i)0.15, (ii)0.70	12
12	Normal Distribution	9740/02	(i) 0.0106 (ii)0.901 (iii)0.274, 306	14